Complex Analysis Homework 1

1) Let $z = \cos\theta + i\sin\theta$.

(i) Show that $z^n + z^{-n} = 2\cos n\theta$, where n is a positive integer.

(ii) Let m be a positive integer. Show that

$$(2\cos\theta)^{2m} = 2\left[\cos 2m\theta + \binom{2m}{1}\cos(2m-2)\theta + \binom{2m}{2}\cos(2m-4)\theta + \cdots + \binom{2m}{m-1}\cos 2\theta\right] + \binom{2m}{m}.$$

(iii) Hence, or otherwise, prove that

$$\int_0^{\frac{\pi}{2}} \cos^{2m}\theta\, d\theta = \frac{\pi}{2^{2m+1}}\binom{2m}{m}$$

where m is a positive integer.

2. Let n be a positive integer.

(a) Assume that θ is not an integral multiple of π.

(i) Prove that

$$\frac{(\cos\theta + i\sin\theta)^n - 1}{\cos\theta + i\sin\theta - 1} = \frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}\left(\cos\frac{(n-1)\theta}{2} + i\sin\frac{(n-1)\theta}{2}\right).$$

(ii) Using the identity $\sum_{k=1}^{n} z^k = z\left(\frac{z^n - 1}{z - 1}\right)$ for $z \neq 1$, or otherwise, prove that $\sum_{k=1}^{n} \cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$ and

$$\sum_{k=1}^{n} \sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}.$$

(iii) Using (a)(ii), or otherwise, prove that

$$\sum_{k=1}^{n} \cos^2(k\theta) = \frac{n}{2} + \frac{\sin n\theta\cos(n+1)\theta}{2\sin\theta}.$$

(b) For $n > 1$, evaluate

(i) $\sum_{k=1}^{n} \sin^2\left(\frac{k\pi}{n}\right)$,

(ii) $\sum_{k=1}^{n} \left(\sin\frac{k\pi}{n} + \cos\frac{k\pi}{n}\right)^2$.

3. Let z be a complex number such that $|z-i|=\sqrt{2}$ and $z \neq \pm 1$. Calculate $Arg\left(\frac{z+1}{z-1}\right)$.

4. Let m be a positive integer.

(a) By using De Moivre's theorem, show that

$$\sin(2m+1)\theta = \binom{2m+1}{1}\cos^{2m}\theta\sin\theta - \binom{2m+1}{3}\cos^{2m-2}\theta\sin^3\theta + \cdots + (-1)^m \sin^{2m+1}\theta.$$

(b) Deduce that the polynomial $p(x) = \binom{2m+1}{1}x^m - \binom{2m+1}{3}x^{m-1} + \cdots + (-1)^m$ has m distinct roots $\alpha_k = \cot^2\left(\frac{k\pi}{2m+1}\right)$ where $k = 1, 2, \ldots, m$.

(c) Prove that $\cot^2\left(\frac{\pi}{2m+1}\right) + \cot^2\left(\frac{2\pi}{2m+1}\right) + \cdots + \cot^2\left(\frac{m\pi}{2m+1}\right) = \frac{m(2m-1)}{3}$.

(d) Show that $\sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{\pi}{2}$. Deduce that

$$\lim_{m\to\infty}\left(\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{m^2}\right) = \frac{\pi^2}{6}$$

5. Suppose that $z = \frac{1}{2}(\cos\theta + i\sin\theta)$.

(a) Show that $1 + z + z^2 + \cdots = \frac{1}{1-z}$.

(b) Express $1 + \frac{1}{2}\cos\theta + \frac{1}{2^2}\cos 2\theta + \frac{1}{2^3}\cos 3\theta + \cdots$ in terms of $\cos\theta$.

(c) Calculate $\arg\left(1 + z + z^2 + \cdots\right)$ for $\theta = -\frac{\pi}{3}$.